

The Role of the Anesthesiologist in Preventing Severe Maternal Morbidity and Mortality

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Abstract: Anesthesiologists are responsible for the safe and effective provision of analgesia for labor and anesthesia for cesarean delivery and other obstetric procedures. In addition, obstetric anesthesiologists often have a unique role as the intensivists of the obstetric suite. The anesthesiologist is frequently the clinician with the greatest experience in the acute bedside management of a hemodynamically unstable patient and expertise in life-saving interventions. This review will discuss (1) risks associated with neuraxial and general anesthesia for labor and delivery, and (2) clinical scenarios in which the obstetric anesthesiologist is commonly called upon to function as a “peridelivery intensivist.”

Key words: anesthesiology, obstetrics, maternal morbidity, cesarean, neuraxial anesthesia

Neuraxial anesthesia and analgesia is considered to be the technique of choice for labor analgesia and cesarean delivery anesthesia. Advantages of neuraxial blocks for labor include superior pain management,^{1,2} availability of an in situ epidural catheter should an urgent (or nonurgent) cesarean delivery be necessary, and decreasing circulating catecholamines,³ of importance for parturients with hypertensive disorders of pregnancy or other cardiac comorbidities.

Neuraxial anesthesia for cesarean delivery is similarly associated with several advantages for cesarean delivery. These include less exposure of the newborn to (maternal) medications,⁴ better early

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maternal and fetal bonding,⁵⁻⁷ and avoidance of general anesthesia. Although in the setting of enhanced supervision, end-tidal CO₂ monitoring, and difficult airway algorithms, the general anesthesia-related maternal mortality has improved to be comparable to neuraxial anesthesia,⁸ there are still several challenges that remain.

General Anesthesia-related Complications, Morbidity, and Mortality

In contemporary surveys, general anesthesia is the mode of anesthesia for 3% to 4% of elective cesarean deliveries, however, it accounts for 14% to 19% of emergency cases.⁹ Common indications for general anesthesia for cesarean delivery include nonreassuring fetal heart rate tracing in a patient without an in situ epidural, cord prolapse, acute placental abruption, inability to achieve adequate anesthesia from a neuraxial technique, or maternal contraindication to neuraxial blockade (eg, coagulopathy, severe thrombocytopenia, or tethered spinal cord). General anesthesia can usually be administered quickly and safely and allows for a delivery that is free of maternal pain and anxiety. However, in rare cases general anesthesia can be associated with severe maternal morbidity and mortality, most notably in the form of difficult or “failed” intubation, pulmonary aspiration, or postoperative pulmonary events.

DIFFICULT OR FAILED INTUBATION

Difficult airway or failed intubation is perhaps the most feared complication related to general anesthesia, in both general and obstetric populations. The risk of “failed” intubation (typically defined as being unable to complete the intubation as planned) is significantly higher in the obstetric population, with a reported incidence in recent studies ranging from 1 in 250 to 1300 intubations.¹⁰⁻¹³

There are many factors that contribute to the obstetric airway being potentially difficult to manage. The Mallampati score has been shown to increase over the course of pregnancy and during labor,¹⁴⁻¹⁶ oral and nasal mucosa becomes edematous, hypervascular, and prone to bleeding, and a combination of decreased functional residual capacity and increased oxygen consumption leads to rapid desaturation when apneic. The emergent nature of cases in which general anesthesia is used for cesarean delivery likely contributes as well, with one survey finding that the relative risk of failed intubation in emergency cesarean delivery was approximately double that in elective cases (relative risk, 1.79; 95% confidence interval, 0.51-5.26).¹⁰ For a parturient with an anticipated difficult airway, the placement of an epidural catheter early in the delivery course can help to avoid the morbidity associated with an urgent general anesthetic. A neuraxial technique is still typically preferred in these women, even for urgent or emergent delivery, as neuraxial block (especially spinal) may take less time than an awake fiber optic intubation and will preserve maternal oxygenation and ventilation. However, in cases where neuraxial techniques are not possible, awake fiber optic intubation or other airway tools should be used to secure the airway, if indicated. Maintaining maternal oxygenation and ventilation remains the most important consideration, even in cases of fetal distress.

In the case of the unanticipated difficult airway, the obstetric anesthesiologist should adhere to the principles of the difficult airway algorithm. The algorithm is based on the same key principles in obstetric patients as in general surgical patients: calling for help, optimizing patient positioning, letting up on cricoid pressure if it is making laryngoscopy more difficult, having the most experienced provider take over laryngoscopy, using a video laryngoscope, and considering

placement of a supraglottic airway such as a laryngeal mask airway (LMA). LMA is the rescue airway of choice, especially in the setting of difficult mask ventilation, and will often allow for oxygenation and ventilation in a patient who could not be intubated. Once an LMA has been placed successfully, the decision about whether to proceed with surgery should be based on assessment of maternal and fetal condition; many cases have been described of cesarean delivery being performed with LMA as the final airway device with good maternal and fetal outcomes.^{17–20}

PULMONARY ASPIRATION

Pulmonary aspiration in the setting of general anesthesia is a controversial topic in obstetric anesthesia. Historically, aspiration was a major cause of morbidity and mortality for parturients—it was the most common cause of mortality related to obstetric anesthesia in the United States before 1990, and during that same time it has been estimated that the risk of death in patients who received general anesthesia for cesarean delivery was 17 times greater than that of patients who received regional anesthesia.²¹ However, in recent decades, pulmonary aspiration has become rare, with contemporary surveys finding an incidence of 0% to 0.46%.^{12,22,23} The sharp decline can be attributed to a combination of factors, including increased awareness of the risk of aspiration among anesthesiologists, use of rapid sequence intubation techniques when general anesthesia is indicated, increased use of neuraxial techniques for cesarean delivery, use of antacids before cesarean delivery, and the widespread practices of having patients fast before planned cesarean delivery, and restricting oral intake to clear liquids during active labor.

Conditions that delay gastric emptying, increase intra-abdominal pressure, or decrease lower esophageal sphincter tone

may place the patient at increased risk of pulmonary aspiration. Such conditions include pregnancy, labor (especially in the presence of epidural or intrathecal opioids),^{24–29} obesity,^{30,31} diabetes mellitus,³² and gastroesophageal reflux disease. Once aspiration occurs, evidence from animal models suggests that the risk and severity of aspiration pneumonitis is inversely related to the pH of the aspirate, and directly related to the volume.^{31,33–36} The consistency of the aspirate is also important—solid matter poses the risk of asphyxiation, and particulate matter is associated with worse outcomes in animal models.³⁵

Although rare, pulmonary aspiration is associated with significant morbidity and mortality when it does occur.^{37–39} The sequelae of aspiration and severity of symptoms depend on multiple factors, including the nature of the aspirate, the patient's baseline health and comorbidities, and how the event is managed in the acute and chronic phases. Most aspiration events that occur in the setting of general anesthesia are witnessed by an anesthesia provider. Initial symptoms include tachypnea (in the spontaneously breathing patient), tachycardia, bronchospasm, and hypoxemia. Cardiac arrest has also been described both in case reports and in animal models.^{33,40} Radiographic signs lag behind clinical symptoms by as much as 24 hours, meaning that chest x-ray is unlikely to be useful in the operating room, however, 85% to 90% of patients will eventually develop alveolar infiltrates in dependent parts of the lung (in mild cases) or diffusely (in severe cases).^{41,42}

Possible sequelae of pulmonary aspiration include acute respiratory distress syndrome, aspiration pneumonia, and death.^{37–39} Management of pulmonary aspiration depends on the patient's presentation. The oropharynx should be suctioned and airway rapidly secured. Rigid bronchoscopy may be considered

if the aspirate contains large pieces of solid food, and the tracheobronchial tree should be suctioned to remove any remaining liquid. Prophylactic antibiotics are likely not necessary, although it is worth noting that the use of antacids may interfere with the bactericidal properties of stomach acid, and place the patient at higher risk of developing aspiration pneumonia.^{43,44} Corticosteroids are frequently administered to patients with aspiration pneumonitis, however, their benefits have never been clearly demonstrated, and some studies suggest that they increase the risk of pneumonia and the length of stay in the intensive care unit.^{43,45} Lung protective ventilation strategies, including low-tidal volumes of 4 to 6 mL/kg and plateau pressures of 30 cm of water or less, have been shown to decrease mortality patients with acute respiratory distress syndrome.⁴⁶

Fasting for 8 hours before planned cesarean delivery is widely accepted, however, the practice of restricting oral intake during active labor, or during labor epidural analgesia, varies considerably from institution to institution, and has been challenged in recent years.^{47,48} Arguments for liberalizing oral intake during labor include the low incidence of aspiration in modern obstetric anesthesia and the high energy demands of labor. However, several well-designed, randomized studies and meta-analyses of restrictive versus liberal oral intake during labor have found no difference between groups with regards to important outcomes such as the duration of labor, neonatal outcomes, or the need for cesarean or operative vaginal delivery.^{47,49,50} In light of the seemingly limited consequence of keeping women fasted during labor, and the potential morbidity and mortality associated with pulmonary aspiration, policies of restricted oral intake during labor should probably be maintained.

Neuraxial Anesthesia-related Complications, Morbidity, and Mortality

HEADACHE

Postpartum headache is extremely common, occurring in as many as 40% of postpartum women.^{51,52} Because of the well-known association of neuraxial anesthesia and postdural puncture headache, the anesthesiologist is often asked to evaluate these patients. Single institution retrospective studies have found that ~16% of postpartum headaches are likely to be of postdural puncture origin,⁵¹ related to either intentional dural puncture during a spinal anesthetic procedure, or inadvertent dural puncture during an epidural anesthetic procedure. The overall risk of obstetric postdural puncture headache is 1% to 2%. Postdural puncture headache often presents first with neck stiffness in the absence of fever, and then progresses to a headache that is exacerbated in the upright position.⁵³ These headaches, a symptom of low cerebrospinal fluid pressure, do eventually remit spontaneously, but can often cause considerable maternal discomfort and disruption in caring for her newborn. Pharmacologic (conservative) therapy has been shown to have little or no consistent benefits (eg, caffeine, fioricet, non-steroidal anti-inflammatory drugs, adrenocorticotrophic hormone, cosyntropin)⁵⁴ and overuse of agents such as non-steroidal anti-inflammatory drugs and fioricet can be associated with chronic headache.^{55,56} Autologous lumbar epidural blood patch can improve the headache in many women with many women with post-dural puncture headache, although 3% to 28% of women will require a second blood patch, depending on the study.^{57,58}

Whereas most other postpartum headaches are migraine or tension-type, some are associated with more morbid diagnoses, especially if they are accompanied by coexisting neurological deficits. Headaches that are worse in the supine position (the reverse of many women with post-dural puncture

headache) are particularly ominous, as they may be associated with increased intracranial pressure. Albeit a low-incidence event (occurring in about 34/100,000 deliveries),⁵⁹ stroke is a leading cause of maternal mortality.⁶⁰ Postpartum women, particularly those that are preeclamptic, are increasingly at risk for stroke. Other neurological urgencies include cerebral venous sinus thrombosis (with or without stroke), reversible cerebral vasoconstriction syndrome, often with accompanying posterior reversible encephalopathy syndrome. Although rare in a young otherwise healthy obstetric patient, meningitis can occur, as can headaches from tumor, medications, illicit substances, and other etiologies.⁶¹ Being well versed in the differential diagnosis of postpartum headache and able to discern, which patients need urgent neurological expert attention, the anesthesiologist can play a key role in mitigating related maternal morbidity and mortality.

HIGH OR TOTAL SPINAL

High spinal refers to progression of block height from an intrathecal injection to a more cephalad level than was intended, with variable consequences depending on the degree of spread. "Total spinal" is a term that is colloquially used to refer to a high spinal block that results in loss of consciousness. It is important to note that the loss of consciousness is usually a result of hypotension causing hypoperfusion of the brainstem, as opposed to the direct effects of local anesthetic on the brain.⁶² Furthermore, the effects of an inappropriately high block are not necessarily immediate, and can continue to evolve over several minutes from the time of injection. Therefore, the distinction between the 2 (high vs. total spinal) is primarily one of semantics, and the providers should anticipate progression to these symptoms when a high spinal is suspected. For the purposes of this chapter, the term high spinal will be used to describe both entities.

The reported incidence of high spinal varies widely, ranging from 1 in 16,200 to

1 in 1400.^{22,63–65} It can occur in a variety of clinical situations. The level of spread of local anesthetic is primarily governed by the dose^{66,67} and baricity^{68,69} of the injectate. Spinal anesthetics (eg, for cesarean delivery) performed with low dose, hyperbaric solutions of local anesthetic are less likely to result in a clinically significant degree of high spinal blockade. High spinal is more likely to result from an epidural catheter that has been inadvertently placed, or has migrated into, the intrathecal or subdural compartment.

High spinal may occur include during the administration of a "test dose" at initiation of a newly placed epidural catheter, while bolusing an epidural catheter for breakthrough pain or to convert to a surgical anesthetic, or when a spinal anesthetic is attempted in a patient after epidural block. The purpose of the epidural test dose is to identify either intrathecal or intravenous placement of an epidural catheter, and therefore avoid subsequent inadvertent injection of local anesthetic into the cerebrospinal fluid or a blood vessel. The typical test dose contains 45 mg of isobaric lidocaine and 15 µg of epinephrine. It is important to note that onset of spinal anesthesia following 45 mg of lidocaine is variable, and it may take in excess of 5 minutes to see clinically significant sensory and motor blockade.⁷⁰ If additional local anesthetic is injected into the epidural catheter in the first couple of minutes following an apparently negative test dose, high spinal block could ensue. Aspiration of the epidural catheter is not 100% sensitive for detecting intrathecal catheter placement.^{71,72}

Conversion of labor epidural catheter to a surgical anesthetic for cesarean delivery is always a high-risk period. Such transitions are frequently marked by urgency due to deterioration of either the fetal or maternal condition, which compels the anesthesiologist to inject large volumes of high-concentration local anesthetic into the epidural catheter over a

short period of time. These injections may occur as the patient is being transferred from the labor room to the operating suite, often without monitoring. As many institutions now use very low concentrations of local anesthetic for labor epidural analgesia, it is possible that an intrathecal or intravascular catheter could be unrecognized during labor, thereby exposing the patient to severe morbidity when dosing the epidural for surgical anesthesia for cesarean delivery. In cases where the labor epidural fails to convert to an adequate surgical anesthetic, proceeding with a spinal anesthetic must be done with extreme caution, as this sequence is high risk for the development of a high spinal.^{22,73,74}

Regardless of the etiology, signs, symptoms, the management of high spinal are the same. Symptoms depend on the degree of cephalad spread of the block. Sympathectomy and involvement of cardiac accelerator fibers from T1-T4 cause severe hypotension and bradycardia. Profound hypotension can result in nausea, confusion, agitation, or loss of consciousness. Hypopnea and hypoxia can be caused by either weakness of the accessory muscles of breathing, or by diaphragmatic paralysis once the block reaches the C3-5 nerve roots. Respiratory depression may be presaged by voice weakness or inability to speak. Fetal condition may also be compromised secondary to hypotension or hypoxia. Treatment includes placing the patient in a modest Trendelenburg position with left uterine displacement to decrease venous pooling, and further supporting hemodynamics with volume expansion, vasopressors, or inotropes. There is a high risk of aspiration secondary to decreased level of consciousness, supplemental oxygen should be administered and the airway should be secured early, although in some instances it may be sufficient to support respiration with bag mask ventilation. The fetal heart rate should be monitored during this time,

bearing in mind that fetal condition may stabilize once maternal hypotension and hypoxia are addressed. If fetal bradycardia persists once the maternal condition has been stabilized, delivery may be indicated.

Subdural injection of local anesthetic is a distinct clinical entity, and can also result in inappropriately high levels of anesthesia. The reported incidence ranges from 0.82% to as high as 11%.^{75,76} Subdural injection occurs when local anesthetic is injected between the dura mater and the arachnoid mater, and is characterized by a high, patchy block sensory block, frequently with sacral sparing and relatively preserved motor function. It may also present as inadequate epidural analgesia or anesthesia.⁷⁶ Cranial nerves may be involved, and Horner syndrome has been reported.⁷⁷ Hypotension is typically less profound than with a high spinal, however, apnea and loss of consciousness can occur secondary to direct intracranial involvement. Onset of the block tends to be slower and more insidious than that seen with intrathecal injection, resulting in a delayed diagnosis. Treatment is symptomatic and depends on the degree and quality of resulting block.

INTRAVASCULAR INJECTION OF LOCAL ANESTHETIC

Intravascular injection of local anesthetic and subsequent local anesthetic systemic toxicity (LAST) is a rare but potentially devastating complication of epidural analgesia during labor or for cesarean delivery. Ideally, intravascular placement or migration of an epidural catheter will be identified before administration of large doses of local anesthetic. An epinephrine-containing test dose has repeatedly been demonstrated to have excellent sensitivity for detection of an intravascular catheter in non-beta-blocked patients.^{78,79} Unfortunately, due to maternal heart rate variability during labor, the specificity may be considerably lower.^{78,80} The potential for false positives combined with

concerns about the effects of epinephrine on placental blood flow in the setting of a positive test dose have led some institutions to abandon the test dose in obstetrics entirely. Although this practice is controversial, there are certain situations in which it is advisable to avoid an epinephrine-containing test dose, such as in patients with severe hypertension, or in the setting of certain forms of heart disease in which a sudden increase in afterload may be detrimental (eg, cardiomyopathy with reduced ejection fraction or severe regurgitant valvular lesions). Aspiration of the catheter before injection, while highly specific, has been demonstrated to have poor sensitivity.^{81,82} The sensitivity of aspiration may be improved using a multiorifice catheter, but false negatives can still occur.^{83,84} Use of a flexible epidural catheter appears to decrease the risk of erosion into a blood vessel.⁸⁵

Classically, signs and symptoms of LAST start with neurological symptoms such as tinnitus, perioral numbness, agitation, and seizures. Cardiovascular symptoms including hypotension, dysrhythmias, and cardiac arrest typically follow. High plasma levels of bupivacaine may cause cardiotoxicity simultaneously with the onset of neurological symptoms, and can be especially difficult to treat,⁸⁶ as a result of which high-concentration bupivacaine for is no longer used for obstetric anesthesia. Conversely, 2-chloroprocaine is rapidly metabolized by plasma cholinesterases, meaning that even if intravascular injection were to occur, the resulting symptoms would be expected to be less severe. Treatment of LAST includes depends upon symptoms, but includes:

- (1) Stopping administration of local anesthetic and calling for help.
- (2) Management of seizure activity if present (starting dose of 2 to 5 mg midazolam). Phenytoin should be avoided as it can worsen cardiotoxicity.
- (3) Administration of 100% oxygen and early tracheal intubation (cardiotoxicity is exacerbated by hypoxia, hypercapnia, and acidemia).
- (4) Support of hemodynamics including placing the patient in left uterine displacement, initiation of advanced cardiac life support, and early administration of 20% lipid emulsion (1.5 mL/kg over 1 min followed by 0.25 mL/kg/min, bolus may be repeated for persistent cardiovascular collapse, maximum dose of 10 mL/kg).⁸⁷ For refractory collapse, cardiovascular surgery should be consulted for possible initiation of cardiopulmonary bypass or extracorporeal membrane oxygenation.
- (5) Avoidance of high-dose epinephrine (> 1 mg/kg), vasopressin, calcium channel blockers, beta blockers, and local anesthetics.
- (6) Cesarean delivery should be considered early as it may improve efforts at maternal resuscitation.

EPIDURAL HEMATOMA

Hemorrhage into the epidural space is an exceedingly rare complication of epidural for labor or delivery, albeit 1 that is associated with severe maternal morbidity. It is difficult to accurately estimate the incidence of epidural hematoma, as it is so rare that few events are seen even in very large surveys of labor epidurals, however, the reported incidence in recent studies ranges from 1 in 160,000 to 1 in 250,000.^{22,88,89} Epidural hematoma is rarer in the obstetric population than in the general surgical population.^{88,90} The most important risk factors in the obstetric population seem to be disorders of coagulation (including the iatrogenic administration of anticoagulants) and thrombocytopenia, especially when associated with the syndrome of hemolysis, elevated liver enzymes, and low platelets (HELLP syndrome),^{88,91,92} although there are very few reported cases in the

literature. Other risk factors include multiple or traumatic placement attempts.^{91,92}

There is no data to definitively determine the minimal platelet count required to safely perform a neuraxial anesthetic. Epidural hematoma can occur in the setting of a normal platelet count,⁹³ and neuraxial anesthetics have been performed uneventfully in patients who have subsequently been discovered to have profound thrombocytopenia.^{94–96} Most anesthesiologists will place an epidural block in patients with a platelet count of > 70 to 80,000 without other contraindications. Risk of epidural hematoma in patients with a platelet count of 70 to 100,000 has an upper 95% confidence interval of 0.2%.⁹⁶ Factors that may influence the decision include the etiology of the thrombocytopenia, the rapidity of the drop in platelet count, whether or not the patient has a reassuring airway examination, and the risk of general anesthesia. Single shot spinal technique is generally considered lower risk than placement of an epidural catheter due to the smaller needle size, relative ease of technique, and lack of need to later remove a catheter from the epidural space. The decision about when to proceed with a neuraxial anesthetic in the setting of thrombocytopenia requires a frank and thorough discussion between providers and their patients about the risks and benefits associated with the choices, so that a truly informed decision can be made.

Increasingly, parturients may be receiving on anticoagulants at the time of labor or for cesarean delivery. Iatrogenic anticoagulation is a potential risk factor for epidural hematoma^{97,98} and the appropriate timing of discontinuation or transition to shorter acting agents requires careful interdisciplinary planning. Depending on the agent being used, decision to proceed with neuraxial anesthetic may be based on timing of drug administration, laboratory values, or both. The Society of Obstetric Anesthesiology and

Perinatology recently created a multidisciplinary consensus statement on the care of pregnant and postpartum patients on anticoagulants and placement of a neuraxial block.⁹⁹ Multidisciplinary planning and communication between anesthesiologists, obstetricians, nurses, and patients will be key in the effort to provide safe and effective care to this growing population of patients.

The symptoms of epidural hematoma, caused by compression of neural structures and resultant ischemia, include pain in the back and legs, loss of sensation in the lower extremities, motor weakness, and loss of bowel or bladder control. Treatment is usually surgical, often with laminectomy to evacuate the hematoma. Early recognition, appropriate imaging, and prompt neurosurgical intervention are vital to improve outcomes—1 review of 30 cases of epidural hematoma showed that patients who received surgery within 12 hours of diagnosis had improved outcomes as compared with patients with similar preoperative neurological deficits in whom surgery was delayed.¹⁰⁰ Peripheral nerve injury after delivery is quite common but fortunately is usually benign and self-limited¹⁰¹; the high incidence of these types of symptoms may obscure or delay diagnosis. Patients with signs or symptoms of peripartum neurological injury require a detailed history and physical examination, and frequent follow-up to determine the time course and evolution of their symptoms.

EPIDURAL ABSCESS AND MENINGITIS

Infection in neuraxial space is a rare complication of neuraxial anesthesia on labor and delivery, although with a reported incidence of epidural abscess ranging from 1 in 506,000¹⁰² up to 1 in <5000,^{103,104} it appears to be more common than epidural hematoma. The incidence of postspinal meningitis is difficult to estimate because it is not consistently reported in surveys of complications

related to obstetric anesthesia. In the ASA Closed Claim Projects database from 1980 to 1999, neuraxial infections (including epidural abscess and meningitis) were the most common cause of neuraxial injury in obstetric cases.¹⁰⁵ It is important to note that epidural abscess has been reported to occur spontaneously in both obstetric and nonobstetric patients,^{106,107} and that community-acquired meningitis can occur in pregnancy or the postpartum period.

In the setting of a neuraxial anesthetic, epidural catheterization is a prerequisite for epidural abscess formation; no cases have been reported in patients who received spinal anesthesia alone.¹⁰⁸ Possible risk factors for epidural abscess include prolonged catheterization, difficult or traumatic procedure at the time of initial insertion, administration of epidural opioids without local anesthetic, and patient-level factors such as diabetes or immunosuppression.^{104,109–113} *Staphylococcus aureus* is the most common pathogen, and skin the most likely source.¹⁰⁶ In the case of meningitis, dural puncture is likely a prerequisite for infection to occur.^{108,114,115} *Streptococcus salivarius* is the most common pathogen, and possible risk factors include poor sterile technique, infection at another site (eg, chorioamnionitis), prolonged catheterization, labor, and diabetes or immunosuppression.^{104,108,109,113} Evidence for some of these risk factors is based on the fact that postspinal meningitis is exceedingly rare in the setting of planned cesarean deliveries, which typically take place before labor, and in the more sterile environment of the operating room, where providers are more likely to wear a mask and adhere to rigorous sterile technique.

With regards to preventing neuraxial infections, sterile technique is the most significant modifiable risk factor for the anesthesiologist. Wearing a mask is of particular importance as there is data to

suggest that anesthesiologists may neglect this and other aspects of sterile technique.^{116–119} Skin preparation and adequate drying time are also important. There is growing evidence that chlorhexidine in 70% alcohol is superior to povidone-iodine for eliminating bacteria on the skin.^{103,120,121} The largest concentration of bacterial colonies on the skin exists in hair follicles,¹²⁰ emphasizing the importance of adequate duration of preparation to penetrate these areas.¹²¹ A sterile, occlusive dressing should be applied over the insertion site, and maintained until the catheter is removed. Full guidelines regarding use of aseptic or sterile technique can be found in the American Society of Anesthesiologists' Practice Advisory for the Prevention, Diagnosis, and Management of Infectious Complications Associated with Neuraxial Techniques.¹²²

As with other space-occupying lesions of the vertebral canal, the symptoms of epidural abscess relate to compression and ultimate ischemia of the spinal cord. Initial symptoms include back pain (with or without radiation), localized tenderness over the insertion site (with or without drainage), neck stiffness, and fever. Only a small portion of patients will have the classic triad of back pain, fever, and neurological symptoms^{123,124}; a high index of clinical suspicion is required to avoid diagnostic delays, which can result in poor outcomes. Laboratory evaluation typically reveals leukocytosis and increased c-reactive protein. If untreated, symptoms may progress to include saddle anesthesia, leg weakness, incontinence of the bowel and bladder, and other symptoms of cauda equina syndrome. If epidural abscess is suspected, an magnetic resonance imaging should be obtained to confirm the diagnosis, and prompt neurosurgical decompression is indicated if neurological symptoms are already present. If symptoms are mild and neurological sequelae are absent, there may be a

role for nonsurgical management with antibiotics and close follow-up.^{125,126} In either case, early and appropriate antibiotic therapy is important. Although many patients make a full neurological recovery, some will have permanent neurological deficits, and epidural abscess can also lead to sepsis and death.¹²⁷ Delayed diagnosis and increased duration of neurological symptoms conveys a higher likelihood of permanent disability.¹²⁴

The classic triad of symptoms of acute bacterial meningitis is fever, nuchal rigidity, and altered mental status. A severe, bilateral headache is also common. Not all patients present with the classic triad, although most have at least 2 of the 4 most common symptoms.^{128,129} There is significant overlap between the symptoms of bacterial meningitis and those of spinal epidural abscess. Clinical risk factors such as placement of an epidural catheter, and known puncture of the dura should be considered, and imaging should be obtained early to rule out epidural abscess. When bacterial meningitis is suspected, appropriate antibiotics should be administered early; a delay in initiating antibiotics is associated with worse outcomes, including death.¹²⁹ Obstetric patients who receive early diagnosis and therapy are expected to make a full recovery.¹⁰⁶

The Role of the Anesthesiologist in Obstetric Crises

The primary threats to maternal well-being in the peripartum period are not related to anesthesia, but rather stem from obstetric crises and underlying maternal disease. From 2011 to 2013, anesthetic complications accounted for 0.2% of pregnancy-related deaths in the United States.¹³⁰ The leading causes of pregnancy-related deaths in that time period were cardiovascular disease (accounting for 15.5% of pregnancy-related deaths), noncardiovascular medical disease (14.5%), infection or sepsis

(12.7%), hemorrhage (11.4%), cardiomyopathy (11.0%), thrombotic pulmonary embolism (9.2%), hypertensive disorders of pregnancy (7.4%), cerebrovascular accidents (6.6%), and amniotic fluid embolism (5.5%). Regardless of the etiology of the crisis, the obstetric anesthesiologist is frequently called upon when a patient becomes unstable, and offers a perspective and skill set that is unique on the labor and delivery unit. The risk factors, presentation, and management of each of the obstetric and peripartum emergencies are covered in detail in other articles in this issue. However, there are certain common themes to the skills and services that the obstetric anesthesiologist can provide in managing these high-risk patients.

AIRWAY MANAGEMENT

Obstetric patients sometimes require ventilatory support or tracheal intubation outside of the operating room. Any patient with an altered level of consciousness, impaired ability to protect her airway, or respiratory distress may require airway management. Emergency airway equipment should be readily available in the obstetrics unit.

INVASIVE MONITORS AND VASCULAR ACCESS

Invasive blood pressure monitoring may be indicated in any patient with hemodynamic instability. Examples of patients who may benefit from an arterial catheter are parturients with cardiovascular, valvular heart disease, or cardiomyopathy, postpartum hemorrhage, and refractory hypertension. An arterial line is often useful in these same settings to facilitate frequent blood draws when following serial laboratory values. Although more invasive and a slightly higher risk procedure, central venous access is sometimes indicated in patients with difficult intravenous access, an ongoing need for vasoactive infusions, or to monitor central venous pressure.

BLOOD PRODUCT ADMINISTRATION

Many labor and delivery units have protocols in place for the massive transfusion and the management of obstetric hemorrhage. Anesthesiologists are familiar with the equipment and processes of safely and effectively administering blood products, and are natural leaders in this area.

TRANSTHORACIC ECHOCARDIOGRAPHY (TTE)

Bedside TTE can be useful in a variety of settings, most notably in the assessment of the hemodynamically unstable patient. TTE is especially useful for assessing volume status and adequacy of resuscitation, identifying major valvular abnormalities, assessing gross ventricular function, and identifying pericardial effusions.¹³¹ Bedside TTE has its limitations, and to avoid the pitfalls of underdiagnosis or overdiagnosis, any unexpected findings should be confirmed with formal echocardiography.

Conclusion

Techniques for the provision of labor analgesia for vaginal delivery and anesthesia for cesarean delivery are remarkably safe. However, it is important for all members of the care team to be aware of the rare complications that can attend these interventions, and to be prepared to act as first responders to prevent those complications from progressing to cause severe maternal morbidity or mortality. Healthcare providers in the obstetric unit should also recognize the unique training and skill set that anesthesiologists possess that allow them to manage a range of obstetric crises, from severe hypertension to hemorrhage. Labor and delivery units should develop protocols to ensure that they call on anesthesiologists to assist in the manage these crises when they occur.

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